Digital Stethoscope



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1. Overview

A **Digital Stethoscope** captures internal body sounds (heartbeat, lungs, etc.) using a sensor and processes them digitally for output to headphones, mobile apps, or cloud platforms for telemedicine.

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2. Key Features

- Heartbeat & Lung sound detection
- Audio filtering & amplification
- OLED/Touch display for heart rate display
- Bluetooth or Wi-Fi for wireless data
- Mobile App (optional)
- Data Logging
- Rechargeable battery-powered

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3. Block Diagram

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```
[Microphone/Piezo Sensor]
          ↓
  [Pre-Amplifier Circuit]
          ↓
  [Low-Pass/Bandpass Filter]
          ↓
  [ADC Input to Microcontroller]
          ↓
[Microcontroller with DSP (e.g., STM32/ESP32)]
     ↓                ↓              ↓
[OLED Display]   [Audio Output]   [Bluetooth/Wi-Fi]
                                      ↓
                          [Mobile App/PC Viewer]
```

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4. Hardware Architecture

🔹 Sensor Input

- **Piezoelectric Disc or Electret Condenser Microphone**: Captures low-frequency body sounds.

🔹 Analog Front-End

- **Pre-Amplifier (e.g., LM358 Op-Amp)**: Boosts weak signal from sensor.
- **Filter Circuit**: Low-pass filter (~150 Hz) to remove noise.

🔹 ADC + DSP Processor

- **Microcontroller**:
    - **ESP32** (for Wi-Fi/Bluetooth & audio processing)
    - Or **STM32F103C8T6** (for better signal processing)

🔹 Audio Output

- **3.5 mm Jack or Bluetooth Audio**
- Optional: Use DAC for cleaner analog output.

🔹 Display Interface

- **OLED 0.96" or TFT 1.8" Display**
- Shows waveform, BPM, and battery status.

🔹 Wireless Interface (optional)

- **Bluetooth Module (HC-05) or ESP32's in-built BT**
- For streaming to smartphone.

🔹 Power Management

- **3.7V Li-ion Battery**
- **TP4056 Charging Module**
- **Boost Converter (MT3608)** for 5V output

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5. Components List

Component	Specification/Part Number
Microcontroller	ESP32 Dev Board / STM32F103
Sensor	Electret Microphone/Piezo Sensor
Pre-Amp IC	LM358 Op-Amp
Filter Capacitors/Resistors	As per design (10k, 100nF, etc.)
Display	OLED 0.96” I2C / TFT Display
Audio Output	3.5mm Jack or Bluetooth Audio
Battery	3.7V 1000mAh Li-ion
Charger Module	TP4056 with protection
Boost Converter	MT3608 5V Step-up Module
Enclosure	3D-Printed/Plastic Case
PCB	Custom-Designed 2-layer PCB

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6. Software Design

Development Environment

- **Arduino IDE** (for ESP32/STM32 with STM32duino)
- **PlatformIO** (optional for advanced users)

Features

- ADC sampling of body sound signals
- Apply **Band-pass filtering in software (20–200 Hz)**
- Display waveform and calculated BPM
- Output sound via DAC (or PWM) to headphones
- Optional: Send real-time data via Bluetooth/Wi-Fi

Example Libraries

- `Adafruit_GFX.h` and `Adafruit_SSD1306.h` (for OLED)

- `BluetoothSerial.h` (for ESP32)
- `FFT.h` (for signal processing)

Optional Mobile App

- Developed using **MIT App Inventor**, **Flutter**, or **React Native**
- Shows BPM, waveform, historical data

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7. PCB Design Considerations

- Use ground planes to reduce noise
- Isolate analog and digital grounds
- Shield analog paths if possible
- Use 0805/0603 SMD parts for compact design
- Keep analog signal path short

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8. Enclosure Design

- 3D-printed case with:
    - Sensor chamber
    - Display window
    - Audio jack and charging port
- Use flexible diaphragm to enhance pickup

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9. Testing & Calibration

- Use recorded heartbeat audio for initial testing
- Calibrate gain in software for different sensor strengths
- Compare readings with standard stethoscope or mobile apps

10. Future Enhancements

- AI-based sound classification (murmurs, wheezing)
- Cloud sync for remote diagnosis
- Integration with Electronic Health Records (EHR)
- Multiple sensors for multi-point auscultation

Sample Image Mockup

Would you like me to generate a **realistic product image** for your digital stethoscope, including the display, casing, and wireless features?
